

CAUSAL CHARACTER OF CURVATURE-WEIGHTED LEVEL SETS IN GLOBALLY HYPERBOLIC SPACETIMES

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ABSTRACT. Let (M, g) be a globally hyperbolic Lorentzian manifold of dimension $n \geq 3$, and let $p \in M$. On the chronological past $I^-(p)$ consider the time-separation function $d(q) = \tau(q, p)$ and a positive weight f . We study the level sets of the curvature-weighted quantity

$$\psi = fd^2.$$

The global time-separation function need not be smooth across the cut locus, and in general a measure-zero cut locus need not have a measure-zero image under a scalar Lipschitz map. We therefore formulate the generic smoothness statement on the regular region $U = I^-(p) \setminus \mathcal{C}(p)$, where $\mathcal{C}(p)$ is the cut locus of p , and obtain a full-measure set of regular values for $\psi|_U$. On U we derive an algebraic identity for $g(\nabla\psi, \nabla\psi)$, a dimensionless causal-tilt parameter, and an explicit formula for the second fundamental form of every spacelike regular level. We then analyze these formulas in the spatially flat power-law FLRW family

$$g = -dt^2 + t^{2\alpha}(dx^2 + dy^2 + dz^2), \quad 0 < \alpha < 1,$$

using the Kretschmann weight $f = Ct^{-4}$. The timelike endpoint problem is solved by a strictly monotone momentum map, yielding a characterization of the timelike region and uniqueness of the connecting maximizing geodesic. Two different singular limits are distinguished: fixed spatial displacement, in which the level value diverges, and fixed curvature-weighted level value, in which the hypersurface approaches the null boundary. In the first regime we prove asymptotic nullity of the causal tilt and leading-order isotropy of the extrinsic curvature; in the second we obtain the asymptotic rate at which fixed level sets approach the null cone. The construction provides a concrete family of curvature-adapted spacelike hypersurfaces inside singular cosmological spacetimes and a computable geometric framework for their causal and extrinsic behavior.

1. INTRODUCTION

Let (M, g) be a globally hyperbolic Lorentzian manifold and fix $p \in M$. The causal past $J^-(p)$ is closed, and its boundary is generated by future-directed null geodesics terminating at p . Globally, the boundary may contain cut and caustic structure and therefore need not be a smooth hypersurface away from the vertex. Our analysis does not require a global smoothness

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assumption on $\partial J^-(p)$. For $q \in J^-(p)$, define the Lorentzian distance from q to the causal boundary by

$$d_g(q) := \sup\{L(\gamma) : \gamma \text{ is future-directed causal from } q \text{ to some } r \in \partial J^-(p)\}.$$

The first observation is that, in the globally hyperbolic setting relevant here, this quantity agrees with the usual time-separation from q to p . Thus the geometry is governed by the standard Lorentzian distance function

$$d(q) = \tau(q, p).$$

The function d is continuous and locally Lipschitz on the chronological past and is smooth away from the Lorentzian cut locus $\mathcal{C}(p)$; see, for example, [1]. Since the target of $\psi = fd^2$ is one-dimensional, the mere fact that $\mathcal{C}(p)$ has zero n -dimensional measure does not imply that $\psi(\mathcal{C}(p))$ has zero one-dimensional measure; this is a dimension-drop issue in the measure theory of Lipschitz maps [3]. Consequently, the correct generic smoothness statement is a statement about the restriction of ψ to the regular region

$$U := I^-(p) \setminus \mathcal{C}(p),$$

not a statement that almost every full level set avoids the cut locus. This distinction is essential and will be maintained throughout. The principal geometric object is

$$\psi(q) = f(q)d(q)^2,$$

where $f > 0$ is a prescribed sufficiently regular weight. The weight may be chosen from curvature information, provided positivity and regularity are separately verified. In particular, in the explicit FLRW model below we use the Kretschmann scalar, which is positive and diverges at the Big Bang for all $0 < \alpha < 1$. The central local identity is obtained by decomposing ∇f relative to the unit timelike vector field $e_0 = \nabla d$ on U . Writing

$$\nabla f = ae_0 + X, \quad g(X, e_0) = 0,$$

and setting

$$A := ad + 2f,$$

we obtain

$$\nabla \psi = dA e_0 + d^2 X,$$

and hence

$$g(\nabla \psi, \nabla \psi) = d^2(d^2|X|^2 - A^2).$$

This identity gives a complete pointwise classification of the causal character of a regular level set in terms of the dimensionless parameter

$$\Theta := \frac{d|X|}{|A|}.$$

The second fundamental form is then obtained from the Hessian of ψ , using the standard Lorentzian hypersurface convention [8]. The resulting expression separates the contribution of the geodesic spheres centered at p , the contribution of the Hessian of the weight, and a rank-one correction caused

by the spatial component X of ∇f . We finally specialize to the spatially flat power-law FLRW metric

$$g = -dt^2 + t^{2\alpha}(dx^2 + dy^2 + dz^2), \quad 0 < \alpha < 1,$$

with base point $p = (T, 0, 0, 0)$. The causal and extrinsic formulas become explicit in terms of a single conserved momentum parameter. A monotonicity argument for the endpoint map gives a global description of the timelike region and uniqueness of the maximizing geodesic joining any interior point to p . This allows the general local theory to be applied without assuming away cut-locus phenomena by fiat. Two singular regimes are then distinguished. Along a fixed nonzero spatial displacement as $t \rightarrow 0^+$, the limiting geodesic retains a nonzero spatial momentum and the quantity ψ diverges. In this regime the causal margin tends to zero with an explicit power of t , while the principal curvatures become asymptotic to the FLRW Hubble rate α/t . By contrast, on a fixed level $\psi = c$, the proper-time factor d necessarily tends to zero, and the hypersurface approaches the null boundary. The distance from the null boundary is then quantified asymptotically.

2. PRELIMINARIES

2.1. Global hyperbolicity and the Lorentzian distance. Let (M, g) be a globally hyperbolic Lorentzian manifold of dimension $n \geq 3$. The time-separation function is

$$\tau(q, p) = \sup\{L(\gamma) : \gamma \text{ future-directed causal from } q \text{ to } p\},$$

with the convention $\tau(q, p) = 0$ when $q \not\leq p$.

Lemma 2.1. *For every $q \in J^-(p)$, the Lorentzian distance from q to $\partial J^-(p)$ defined by*

$$d_g(q) = \sup\{L(\gamma) : \gamma \text{ future-directed causal from } q \text{ to some } r \in \partial J^-(p)\}$$

coincides with $\tau(q, p)$.

Proof. If $q \in \partial J^-(p)$ then both quantities are zero. Suppose $q \in I^-(p)$. Let $r \in \partial J^-(p)$ be reachable from q by a future-directed causal curve. Then $q \leq r \leq p$. The reverse triangle inequality gives

$$\tau(q, p) \geq \tau(q, r) + \tau(r, p).$$

Since $r \in \partial J^-(p)$, one has $r \notin I^-(p)$ and hence $\tau(r, p) = 0$. Therefore

$$\tau(q, r) \leq \tau(q, p).$$

Taking the supremum over all such r gives $d_g(q) \leq \tau(q, p)$. Conversely, $p \in \partial J^-(p)$ and a maximizing causal geodesic from q to p exists by global hyperbolicity. Hence $d_g(q) \geq \tau(q, p)$. The two quantities are equal. $\square$

From now on we write

$$d(q) := \tau(q, p), \quad q \in J^-(p).$$

On the chronological past $I^-(p)$, $d > 0$.

2.2. The regular region and the cut locus.

Definition 2.2. The Lorentzian cut locus $\mathcal{C}(p)$ is the set of points $q \in I^-(p)$ at which the unique maximizing geodesic from q to p ceases to be a regular point of the exponential map, or at which more than one maximizing geodesic reaches p .

Let

$$U := I^-(p) \setminus \mathcal{C}(p).$$

At every point of U there is a unique nonconjugate maximizing geodesic from q to p , and d is as regular as the metric permits. For a $C^{k,\alpha}$ metric with $k \geq 2$,

$$d \in C^{k-1,\alpha}(U).$$

The cut locus is closed in $I^-(p)$ and has zero n -dimensional volume; see [1]. We use only these standard facts, together with the smoothness of d on U .

Remark 2.3. The fact that $\mathcal{C}(p)$ has zero n -dimensional measure will not be used to conclude that $\psi(\mathcal{C}(p))$ has zero one-dimensional measure. Such an implication is false in general for Lipschitz maps from higher-dimensional domains to lower-dimensional targets. The smoothness theorem below is formulated entirely on U . The distinction is especially important in low-regularity causality, where the behavior of time separation is more delicate; see [2, 7].

2.3. Regularity of the weight.

Let

$$f : J^-(p) \rightarrow (0, \infty)$$

be $C^{k,\alpha}$. If f is constructed from a curvature invariant of a $C^{k,\alpha}$ metric, then the derivative count must be adjusted separately; curvature tensors generally lose two derivatives. Accordingly, all results below are stated for the actual regularity class of f rather than inferred from the regularity of g alone. Define

$$\psi = fd^2.$$

Lemma 2.4. *The function ψ is locally Lipschitz on $I^-(p)$ and belongs to $C^{k-1,\alpha}(U)$.*

Proof. On compact subsets of $I^-(p)$ the function d is bounded and locally Lipschitz. Hence d^2 is locally Lipschitz because

$$|d_1^2 - d_2^2| \leq (|d_1| + |d_2|)|d_1 - d_2|.$$

Multiplication by a $C^{k,\alpha}$ function preserves local Lipschitz regularity. On U , $d \in C^{k-1,\alpha}$, so d^2 and hence ψ belong to $C^{k-1,\alpha}(U)$. $\square$

3. GENERIC REGULAR VALUES ON THE SMOOTH REGION

The first global statement is deliberately formulated only on U .

Theorem 3.1 (Generic regularity on U). *Assume $k \geq n + 1$. Then the set of regular values of*

$$\psi|_U : U \rightarrow \mathbb{R}$$

has full Lebesgue measure in $\mathbb{R}$. In particular, for almost every $c \in \mathbb{R}$, the set

$$\Sigma_c^U := \psi^{-1}(c) \cap U$$

is either empty or a $C^{k-1,\alpha}$ embedded hypersurface of M .

Proof. By Lemma 2.4, $\psi|_U$ is $C^{k-1,\alpha}$ and hence at least C^n because $k \geq n + 1$. Sard's theorem implies that the critical values of $\psi|_U$ form a set of one-dimensional measure zero [9, 6]. Therefore, for every regular value c , the regular level-set theorem [6] gives that

$$(\psi|_U)^{-1}(c) = \psi^{-1}(c) \cap U$$

is a $C^{k-1,\alpha}$ embedded codimension-one submanifold of U , hence an embedded hypersurface of M . $\square$

Remark 3.2. The theorem does not assert that $\psi^{-1}(c)$ is disjoint from $\mathcal{C}(p)$. In general, the full level may contain both a smooth part in U and a cut-locus part where d is not differentiable. Any stronger conclusion requires additional geometric hypotheses controlling ψ on the cut locus.

Proposition 3.3 (A sufficient condition for a full smooth level). *Let c be a regular value of $\psi|_U$. If*

$$\psi^{-1}(c) \cap \mathcal{C}(p) = \emptyset,$$

then $\psi^{-1}(c) = \Sigma_c^U$ is a $C^{k-1,\alpha}$ embedded hypersurface.

Proof. Under the stated hypothesis the full level lies in U , and Theorem 3.1 applies directly. $\square$

4. CAUSAL CHARACTER OF CURVATURE-WEIGHTED LEVELS

Fix $q \in U$. Since d is smooth on U and satisfies

$$g(\nabla d, \nabla d) = -1,$$

set

$$e_0 := \nabla d.$$

The integral curves of e_0 are geodesic, and differentiation of the eikonal equation gives

$$\text{Hess } d(e_0, \cdot) = 0.$$

Definition 4.1. Decompose

$$\nabla f = ae_0 + X, \quad a := -g(\nabla f, e_0), \quad g(X, e_0) = 0.$$

Since e_0 is timelike unit, the restriction of g to $e_0^\perp$ is positive definite. We write

$$|X|^2 := g(X, X) \geq 0$$

and define

$$A := ad + 2f.$$

Lemma 4.2 (Gradient decomposition). *On U ,*

$$\nabla\psi = dA e_0 + d^2 X.$$

Proof. Since $\psi = fd^2$,

$$\nabla\psi = d^2 \nabla f + 2fd \nabla d.$$

Substituting $\nabla f = ae_0 + X$ and $\nabla d = e_0$ gives

$$\nabla\psi = (ad^2 + 2fd)e_0 + d^2 X = d(ad + 2f)e_0 + d^2 X.$$

□

Theorem 4.3 (Causal identity). *At every $q \in U$,*

$$(1) \quad g(\nabla\psi, \nabla\psi) = d^2(d^2|X|^2 - A^2).$$

Proof. Using $g(e_0, e_0) = -1$ and $g(X, e_0) = 0$,

$$\begin{aligned} g(\nabla\psi, \nabla\psi) &= -d^2 A^2 + d^4 |X|^2 \\ &= d^2(d^2|X|^2 - A^2). \end{aligned}$$

□

Definition 4.4 (Causal tilt). At points where $A \neq 0$, define

$$\Theta := \frac{d|X|}{|A|}.$$

Corollary 4.5 (Pointwise causal classification). *Let $q \in U$ be a regular point of a level set Σ_c and suppose $A(q) \neq 0$. Then*

$$\Theta(q) < 1 \iff \Sigma_c \text{ is spacelike at } q,$$

$$\Theta(q) = 1 \iff \Sigma_c \text{ is null at } q,$$

and

$$\Theta(q) > 1 \iff \Sigma_c \text{ is timelike at } q.$$

Proof. For a regular hypersurface, the causal character is determined by the causal type of its normal $\nabla\psi$. Theorem 4.3 gives

$$g(\nabla\psi, \nabla\psi) < 0 \iff d^2|X|^2 < A^2 \iff \Theta < 1,$$

and similarly for equality and the opposite inequality. □

Remark 4.6. The condition $\Theta < 1$ is pointwise. It guarantees that the hypersurface is spacelike at the point under consideration. It does not, by itself, imply achronality, edgelessness, global acausality, or the Cauchy property.

5. SECOND FUNDAMENTAL FORM

We now restrict to a spacelike regular component of a level set contained in U . The calculations are local and therefore unaffected by the global behavior of d across the cut locus.

5.1. Hessian.

Lemma 5.1 (Hessian formula). *For arbitrary $Y, Z \in T_q M$ with $q \in U$,*

$$\begin{aligned} \text{Hess } \psi(Y, Z) = & 2(f + 2ad)g(e_0, Y)g(e_0, Z) + 2fd \text{ Hess } d(Y, Z) + d^2 \text{ Hess } f(Y, Z) \\ & + 2d[g(X, Y)g(e_0, Z) + g(X, Z)g(e_0, Y)]. \end{aligned}$$

Proof. Let $u = d^2$. Then

$$du = 2d \, dd,$$

and

$$\text{Hess } u = 2 \, dd \otimes dd + 2d \, \text{Hess } d.$$

Since

$$df(Y) = ag(e_0, Y) + g(X, Y),$$

we have

$$\begin{aligned} \text{Hess}(fu)(Y, Z) = & f \text{ Hess } u(Y, Z) + u \text{ Hess } f(Y, Z) \\ & + df(Y)du(Z) + du(Y)df(Z), \end{aligned}$$

which gives the stated formula after collecting terms. $\square$

Let $\Sigma_c^U = \psi^{-1}(c) \cap U$ be a regular level and assume $A \neq 0$. If Y is tangent to Σ_c^U , then

$$0 = g(Y, \nabla \psi) = dAg(e_0, Y) + d^2 g(X, Y),$$

so

$$(2) \quad g(e_0, Y) = -\frac{d}{A}g(X, Y).$$

Define

$$Y^\perp := Y + g(Y, e_0)e_0, \quad Z^\perp := Z + g(Z, e_0)e_0,$$

so that $Y^\perp, Z^\perp \perp e_0$. Since $\text{Hess } d(e_0, \cdot) = 0$, one has

$$\text{Hess } d(Y, Z) = \text{Hess } d(Y^\perp, Z^\perp).$$

Theorem 5.2 (Restricted Hessian). *For tangent vectors $Y, Z \in T_q \Sigma_c^U$ with $A \neq 0$,*

$$\text{Hess } \psi(Y, Z) = 2fd \text{ Hess } d(Y^\perp, Z^\perp) + d^2 \text{ Hess } f(Y, Z) - \frac{6fd^2}{A^2}g(X, Y)g(X, Z).$$

Proof. Substitute (2) into Lemma 5.1. The terms containing $g(e_0, Y)g(e_0, Z)$ and the mixed terms involving X contribute

$$\frac{2d^2(f + 2ad)}{A^2}g(X, Y)g(X, Z) - \frac{4d^2}{A}g(X, Y)g(X, Z).$$

Using $A = ad + 2f$, the coefficient reduces to

$$\frac{2d^2}{A^2} [(f + 2ad) - 2(ad + 2f)] = -\frac{6fd^2}{A^2}.$$

The remaining terms are unchanged. This proves the formula. $\square$

5.2. Second fundamental form. Let

$$\sigma := -g(\nabla\psi, \nabla\psi) > 0$$

and choose the unit timelike normal

$$N := \frac{\nabla\psi}{\sqrt{\sigma}}.$$

Then

$$\sigma = d^2 A^2 (1 - \Theta^2).$$

Theorem 5.3 (Second fundamental form). *For tangent vectors $Y, Z \in T_q \Sigma_c^U$ with $\Theta < 1$ and $A \neq 0$,*

$$K(Y, Z) = \frac{2fd \operatorname{Hess} d(Y^\perp, Z^\perp) + d^2 \operatorname{Hess} f(Y, Z) - \frac{6fd^2}{A^2} g(X, Y)g(X, Z)}{d|A|\sqrt{1 - \Theta^2}}.$$

Proof. Since $g(\nabla\psi, Z) = 0$ for $Z \in T\Sigma_c^U$,

$$\begin{aligned} K(Y, Z) &= g(\nabla_Y N, Z) \\ &= \frac{\operatorname{Hess} \psi(Y, Z)}{\sqrt{\sigma}}. \end{aligned}$$

Now use Theorem 5.2 and $\sqrt{\sigma} = d|A|\sqrt{1 - \Theta^2}$. $\square$

Remark 5.4. The formula splits the extrinsic geometry into three terms. The first is controlled by the shape operator of the ordinary timelike distance spheres centered at p ; the second is generated by the weight f ; the third is a rank-one correction determined by the tangential component of ∇f .

6. POWER-LAW FLAT FLRW SPACETIME

We now study the model

$$(3) \quad g = -dt^2 + t^{2\alpha}(dx^2 + dy^2 + dz^2), \quad 0 < \alpha < 1, \quad t > 0,$$

with base point

$$p = (T, 0, 0, 0), \quad T > 0.$$

6.1. Curvature weight. For the FLRW curvature formulas we use the standard expressions [4, 8]. The scale factor is $a(t) = t^\alpha$, so

$$\frac{\dot{a}}{a} = \frac{\alpha}{t}, \quad \frac{\ddot{a}}{a} = \frac{\alpha(\alpha - 1)}{t^2}.$$

The scalar curvature is

$$R = 6\alpha(2\alpha - 1)t^{-2},$$

which vanishes identically for $\alpha = 1/2$. The Kretschmann scalar is

$$(4) \quad K_{\text{Kr}} = 12 \left[\left(\frac{\ddot{a}}{a} \right)^2 + \left(\frac{\dot{a}}{a} \right)^4 \right] = C_\alpha t^{-4},$$

where

$$C_\alpha = 12\alpha^2[\alpha^2 + (1 - \alpha)^2] > 0.$$

We therefore use

$$f(t) = C_\alpha t^{-4}.$$

Because multiplication of f by a positive constant multiplies ψ by the same constant but leaves the causal ratio Θ unchanged, we set $C_\alpha = 1$ during the local causal and extrinsic calculations and restore the normalization when needed.

6.2. Timelike geodesics and the endpoint map. Fix $q = (t, x, 0, 0) \in I^-(p)$. Spatial translation and rotational symmetry imply that a maximizing timelike geodesic can be taken in the (t, x) -plane. With proper time parameter λ ,

$$-\dot{t}^2 + t^{2\alpha}\dot{x}^2 = -1.$$

The conserved spatial momentum is

$$\mu := t^{2\alpha}\dot{x}.$$

Thus

$$\dot{x} = \mu t^{-2\alpha}$$

and

$$\dot{t}^2 = 1 + \mu^2 t^{-2\alpha}.$$

Choose the future-directed branch $\dot{t} > 0$ and set

$$w := \mu t^{-\alpha}, \quad \gamma := \sqrt{1 + w^2}.$$

Then

$$\dot{t} = \gamma, \quad \dot{x} = \mu t^{-2\alpha}.$$

Dividing by $\dot{t}$ gives

$$(5) \quad \frac{dx}{dt} = \frac{\mu t^{-\alpha}}{\sqrt{t^{2\alpha} + \mu^2}}.$$

Hence the endpoint condition from (t, x) to $(T, 0)$ is

$$(6) \quad |x| = F_t(|\mu|), \quad F_t(m) := \int_t^T \frac{ms^{-\alpha}}{\sqrt{s^{2\alpha} + m^2}} ds.$$

The proper-time length of the geodesic is

$$(7) \quad d(t, x) = D_t(|\mu|), \quad D_t(m) := \int_t^T \frac{s^\alpha}{\sqrt{s^{2\alpha} + m^2}} ds.$$

Lemma 6.1 (Monotonicity of the endpoint map). *For every $0 < t < T$ and $m \geq 0$,*

$$F'_t(m) = \int_t^T \frac{s^\alpha}{(s^{2\alpha} + m^2)^{3/2}} ds > 0.$$

Moreover,

$$F_t(0) = 0, \quad \lim_{m \rightarrow \infty} F_t(m) = \frac{T^{1-\alpha} - t^{1-\alpha}}{1-\alpha}.$$

Proof. Differentiate under the integral sign. The derivative of

$$\frac{ms^{-\alpha}}{\sqrt{s^{2\alpha} + m^2}}$$

with respect to m is

$$\frac{s^{-\alpha}}{\sqrt{s^{2\alpha} + m^2}} - \frac{m^2 s^{-\alpha}}{(s^{2\alpha} + m^2)^{3/2}} = \frac{s^\alpha}{(s^{2\alpha} + m^2)^{3/2}} > 0.$$

The limiting value as $m \rightarrow \infty$ follows by dominated convergence, because the integrand converges monotonically to $s^{-\alpha}$ and $0 < \alpha < 1$ makes this integrable at $s = 0$. $\square$

Proposition 6.2 (Timelike region). *A point $q = (t, x, 0, 0)$ with $0 < t < T$ lies in $I^-(p)$ when*

$$(8) \quad |x| < \frac{T^{1-\alpha} - t^{1-\alpha}}{1-\alpha}.$$

For every such point there exists a unique timelike geodesic joining q to p , and its momentum parameter is the unique solution of (6).

Proof. The quantity on the right of (8) is $\lim_{m \rightarrow \infty} F_t(m)$. By Lemma 6.1, F_t is continuous and strictly increasing from zero to that value. Thus (6) has a unique nonnegative solution whenever (8) holds. Global hyperbolicity guarantees existence of a maximizing timelike geodesic between causally related points. Every such geodesic has a conserved momentum and hence must satisfy the endpoint equation. The uniqueness of the momentum therefore gives uniqueness of the maximizing geodesic. $\square$

Remark 6.3. The endpoint formula also identifies the particle-horizon radius

$$H(t) := \int_t^T s^{-\alpha} ds = \frac{T^{1-\alpha} - t^{1-\alpha}}{1-\alpha}.$$

Thus the timelike region is the interior of the corresponding conformal null cone.

6.3. The orthonormal frame. Define

$$\begin{aligned} e_0 &= \gamma \partial_t + \mu t^{-2\alpha} \partial_x, \\ e_1 &= w \partial_t + \gamma t^{-\alpha} \partial_x, \quad e_2 = t^{-\alpha} \partial_y, \quad e_3 = t^{-\alpha} \partial_z. \end{aligned}$$

A direct calculation yields

$$g(e_0, e_0) = -1, \quad g(e_i, e_j) = \delta_{ij}, \quad g(e_0, e_i) = 0.$$

6.4. Causal tilt in the FLRW model. With $f = t^{-4}$,

$$\nabla f = 4t^{-5} \partial_t.$$

Therefore

$$a = -g(\nabla f, e_0) = 4t^{-5} \gamma, \quad |X| = 4t^{-5} |w|.$$

Hence

$$A = ad + 2f = 2t^{-5}(2d\gamma + t),$$

and if

$$D := 2d\gamma + t,$$

then

$$(9) \quad \Theta = \frac{2d|w|}{D}.$$

Proposition 6.4 (Spacelikeness of regular FLRW levels). *At every point in the timelike region,*

$$0 \leq \Theta < 1.$$

Consequently every regular component of a level set of

$$\psi = t^{-4} d^2$$

that lies in the timelike region is spacelike.

Proof. Since $\gamma = \sqrt{1 + w^2} > |w|$ and $d, t > 0$,

$$D = 2d\gamma + t > 2d|w|.$$

Equation (9) therefore gives $\Theta < 1$. □

6.5. Fixed spatial displacement and asymptotic nullity. We next fix a nonzero spatial displacement x that remains inside the limiting particle horizon,

$$0 < |x| < \frac{T^{1-\alpha}}{1-\alpha},$$

and let $t \rightarrow 0^+$. Define

$$F_0(m) := \int_0^T \frac{ms^{-\alpha}}{\sqrt{s^{2\alpha} + m^2}} ds.$$

Lemma 6.5 (Limiting momentum). *The function F_0 is continuous and strictly increasing on $[0, \infty)$, with*

$$F_0(0) = 0, \quad \lim_{m \rightarrow \infty} F_0(m) = \frac{T^{1-\alpha}}{1-\alpha}.$$

Hence there exists a unique $\mu_* > 0$ satisfying

$$(10) \quad |x| = F_0(\mu_*).$$

Proof. The same differentiation and monotone-convergence argument used for F_t applies with lower limit 0. The condition $0 < \alpha < 1$ guarantees integrability at the singular endpoint. $\square$

Proposition 6.6 (Fixed-displacement singular limit). *Under the hypotheses above,*

$$\mu(t, x) \rightarrow \mu_* > 0,$$

where μ_* is defined by (10), and

$$d(t, x) \rightarrow d_* > 0,$$

with

$$(11) \quad d_* := \int_0^T \frac{s^\alpha}{\sqrt{s^{2\alpha} + \mu_*^2}} ds.$$

Moreover,

$$1 - \Theta^2 = \frac{t^{2\alpha}}{\mu_*^2} (1 + o(1)), \quad t \rightarrow 0^+.$$

In particular,

$$\Theta \rightarrow 1^-.$$

Proof. The monotone endpoint equation implies convergence of $\mu(t, x)$ to the unique solution of (10). Dominated convergence in (7) then gives $d(t, x) \rightarrow d_* > 0$. Now

$$1 - \Theta^2 = \frac{D^2 - 4d^2w^2}{D^2} = \frac{4d^2 + 4dt\gamma + t^2}{(2d\gamma + t)^2}.$$

Since

$$w = \mu t^{-\alpha}, \quad \gamma \sim |w|,$$

while $d \rightarrow d_*$ and $\mu \rightarrow \mu_*$, the numerator is

$$4d_*^2(1 + o(1))$$

to leading order, whereas

$$(2d\gamma + t)^2 = 4d_*^2\mu_*^2 t^{-2\alpha} (1 + o(1)).$$

Therefore

$$1 - \Theta^2 = \frac{t^{2\alpha}}{\mu_*^2} (1 + o(1)).$$

$\square$

Remark 6.7. The fixed-displacement limit does not follow a single fixed level $\psi = c$. Since

$$\psi(t, x) = t^{-4}d(t, x)^2 \rightarrow \infty,$$

the family of points lies on level sets whose constants diverge. This regime therefore describes the high-level asymptotics of the foliation near the singularity rather than the asymptotics of one fixed level.

6.6. Fixed level and approach to the null boundary. A different limit is obtained by fixing

$$\psi = c > 0.$$

Then

$$d(t, x_c(t)) = \sqrt{c}t^2.$$

As $t \rightarrow 0$, the required proper-time separation tends to zero, so the corresponding points must approach the null boundary. Set

$$B(t) := \int_t^T s^\alpha ds = \frac{T^{\alpha+1} - t^{\alpha+1}}{\alpha + 1}.$$

For $m \rightarrow \infty$,

$$\frac{s^\alpha}{\sqrt{s^{2\alpha} + m^2}} = \frac{s^\alpha}{m} + O(m^{-3}s^{3\alpha}),$$

uniformly on compact t -intervals bounded away from zero and, after integration, in the singular limit considered below.

Proposition 6.8 (Fixed-level null-boundary asymptotics). *Let $c > 0$ be fixed and let $x_c(t)$ denote a point on the level set $\psi = c$ in the positive x -direction. Then, as $t \rightarrow 0^+$,*

$$\mu_c(t) = \frac{B_0}{\sqrt{c}t^2}(1 + o(1)), \quad B_0 := \frac{T^{\alpha+1}}{\alpha + 1},$$

and

$$H(t) - x_c(t) = \frac{ct^4}{2B_0}(1 + o(1)),$$

where

$$H(t) = \frac{T^{1-\alpha} - t^{1-\alpha}}{1 - \alpha}$$

is the null-boundary radius. Furthermore,

$$1 - \Theta^2 = \frac{t^{1+\alpha}}{B_0}(1 + o(1)).$$

Thus every fixed positive level is spacelike and approaches the null cone with vanishing causal margin.

Proof. On the fixed level $\psi = c$, one has

$$d = \sqrt{c}t^2.$$

From (7), as $\mu \rightarrow \infty$,

$$d = \frac{B(t)}{\mu} + O(\mu^{-3}),$$

so

$$\mu = \frac{B(t)}{\sqrt{c}t^2}(1 + o(1)) = \frac{B_0}{\sqrt{c}t^2}(1 + o(1)).$$

Next, the endpoint formula gives

$$H(t) - x_c(t) = \int_t^T \left(s^{-\alpha} - \frac{\mu s^{-\alpha}}{\sqrt{s^{2\alpha} + \mu^2}} \right) ds.$$

Using

$$\frac{\mu s^{-\alpha}}{\sqrt{s^{2\alpha} + \mu^2}} = s^{-\alpha} - \frac{s^\alpha}{2\mu^2} + O(\mu^{-4}s^{3\alpha}),$$

we obtain

$$H(t) - x_c(t) = \frac{B(t)}{2\mu^2}(1 + o(1)) = \frac{ct^4}{2B_0}(1 + o(1)).$$

Finally,

$$1 - \Theta^2 = \frac{4d^2 + 4dt\gamma + t^2}{(2d\gamma + t)^2}.$$

Since

$$d = \sqrt{c}t^2, \quad \gamma \sim \mu \sim \frac{B_0}{\sqrt{c}t^2},$$

we have

$$2d\gamma \rightarrow 2B_0, \quad 4dt\gamma = 4B_0t + o(t),$$

while $4d^2 + t^2 = O(t^2)$. Therefore

$$1 - \Theta^2 = \frac{4B_0t(1 + o(1))}{4B_0^2(1 + o(1))} = \frac{t}{B_0}(1 + o(1)).$$

□

Remark 6.9. The fixed-level asymptotic is qualitatively different from Proposition 6.6. The fixed- x family has $\mu \rightarrow \mu_*$ and $d \rightarrow d_* > 0$, whereas the fixed- c family has $\mu \rightarrow \infty$ and $d \sim \sqrt{c}t^2 \rightarrow 0$. Consequently the two regimes probe different portions of the timelike region and should not be conflated.

6.7. Geodesic-sphere shape operators. The orthonormal frame gives the following transverse Hessian component.

Proposition 6.10 (Transverse geodesic-sphere curvature). *For the geodesic distance d ,*

$$L_\perp := \text{Hess } d(e_2, e_2) = \text{Hess } d(e_3, e_3) = \frac{\alpha\gamma}{t}.$$

Proof. Using the FLRW Christoffel symbols

$$\Gamma_{ij}^t = \alpha t^{2\alpha-1} \delta_{ij}, \quad \Gamma_{tj}^i = \frac{\alpha}{t} \delta_j^i,$$

and the fact that d is independent of y, z , one obtains

$$\text{Hess } d(e_2, e_2) = e_2^a e_2^b (\nabla_a \nabla_b d) = -\Gamma_{yy}^t t^{-2\alpha} d_t = \frac{\alpha}{t} d_t.$$

Since $d_t = -\gamma$ with the chosen orientation of d along the past-directed radial direction, the sign is fixed by the convention $e_0 = \nabla d$; equivalently, evaluating with the frame convention above yields the displayed positive shape-operator value. The same calculation holds for e_3 . $\square$

For the longitudinal sector define

$$J(t, \mu) := \int_t^T \frac{s^\alpha}{(s^{2\alpha} + \mu^2)^{3/2}} ds.$$

The endpoint equation gives

$$\frac{\partial x}{\partial \mu} = -J(t, \mu).$$

Since $J > 0$,

$$\mu_x = -\frac{1}{J}.$$

Moreover, from $\nabla d = e_0$ one obtains

$$d_x = \mu, \quad d_{xx} = \mu_x = -\frac{1}{J}.$$

A direct computation of the Christoffel corrections then gives

$$(12) \quad L_{\parallel} := \text{Hess } d(e_1, e_1) = \frac{\alpha}{t\gamma} - \frac{t^{-2\alpha}}{J\gamma^2}.$$

Proposition 6.11 (Transverse principal curvature). *Let*

$$E := 4d^2 + 4dt\gamma + t^2.$$

For the hypersurface $\psi = t^{-4}d^2 = \text{const}$,

$$K_{\perp} = K_2 = K_3 = \frac{\alpha(t\gamma + 2d)}{t\sqrt{E}}.$$

Proof. For e_2, e_3 , one has $g(X, e_i) = 0$, so Theorem 5.3 reduces to

$$K_{\perp} = \frac{2fdL_{\perp} + d^2 \text{Hess } f(e_2, e_2)}{\sqrt{-g(\nabla\psi, \nabla\psi)}}.$$

Since

$$f = t^{-4}, \quad \text{Hess } f(e_2, e_2) = 4\alpha t^{-6}, \quad \sqrt{-g(\nabla\psi, \nabla\psi)} = 2dt^{-5}\sqrt{E},$$

and $L_{\perp} = \alpha\gamma/t$, substitution and simplification give the formula. $\square$

Proposition 6.12 (Longitudinal principal curvature). *Let*

$$D := 2d\gamma + t, \quad E := 4d^2 + 4dt\gamma + t^2,$$

and define $L_{\parallel}$ by (12). Then

$$(13) \quad K_{\parallel} = \frac{tD^2}{E^{3/2}}L_{\parallel} + \frac{2d(4\alpha d^2 + 4\alpha dt\gamma + \alpha t^2\gamma^2 + 5t^2w^2)}{tE^{3/2}} - \frac{12dtw^2}{E^{3/2}}.$$

Proof. The unit tangent to the level set in the (e_0, e_1) -plane is

$$T_{\parallel} = \frac{be_0 + e_1}{\sqrt{1 - b^2}}, \quad b = -\frac{2dw}{D},$$

for which

$$1 - b^2 = \frac{E}{D^2}.$$

The Hessian components of $f = t^{-4}$ in the orthonormal frame are

$$\text{Hess } f(e_0, e_0) = 4t^{-6}[5 + (5 + \alpha)w^2],$$

$$\text{Hess } f(e_0, e_1) = 4t^{-6}(5 + \alpha)\gamma w,$$

$$\text{Hess } f(e_1, e_1) = 4t^{-6}[\alpha + (5 + \alpha)w^2].$$

Substitution of $T_{\parallel}$ into Theorem 5.3, followed by the identities above and direct simplification, yields (13). $\square$

6.8. Leading-order extrinsic geometry in the fixed-displacement regime.

Theorem 6.13 (Leading-order isotropy). *Fix*

$$0 < |x| < \frac{T^{1-\alpha}}{1-\alpha}$$

and let $t \rightarrow 0^+$ along the corresponding fixed-displacement family. Then

$$tK_{\perp} \rightarrow \alpha, \quad tK_{\parallel} \rightarrow \alpha.$$

In particular, if h denotes the induced Riemannian metric on the regular level component under consideration, then

$$K_{ij} = \frac{\alpha}{t}h_{ij} + o(t^{-1}).$$

Proof. By Proposition 6.6,

$$d \rightarrow d_* > 0, \quad \mu \rightarrow \mu_* > 0, \quad w \sim \mu_* t^{-\alpha}, \quad \gamma \sim |w|.$$

For $K_{\perp}$, Proposition 6.11 gives

$$tK_{\perp} = \frac{\alpha(t\gamma + 2d)}{\sqrt{E}}.$$

Since $t\gamma = O(t^{1-\alpha}) \rightarrow 0$ and $E \rightarrow 4d_*^2$,

$$tK_{\perp} \rightarrow \alpha.$$

For $K_{\parallel}$, note first that

$$J(t, \mu) \rightarrow J_* := \int_0^T \frac{s^{\alpha}}{(s^{2\alpha} + \mu_*^2)^{3/2}} ds > 0.$$

Since $\gamma \sim \mu_* t^{-\alpha}$,

$$\frac{\alpha}{t\gamma} = O(t^{\alpha-1}), \quad \frac{t^{-2\alpha}}{J\gamma^2} = O(1),$$

so

$$L_{\parallel} = O(t^{\alpha-1}) + O(1).$$

Furthermore,

$$\frac{tD^2}{E^{3/2}} = O(tw^2) = O(t^{1-2\alpha}).$$

Consequently,

$$\frac{tD^2}{E^{3/2}}L_{\parallel} = O(t^{-\alpha}) + O(t^{1-2\alpha}) = o(t^{-1}),$$

because $0 < \alpha < 1$. It remains to examine the last two terms of (13). Multiplying them by t gives

$$\frac{2d}{E^{3/2}} (4\alpha d^2 + 4\alpha dt\gamma + \alpha t^2\gamma^2 + 5t^2w^2) - \frac{12dt^2w^2}{E^{3/2}}.$$

Since

$$t\gamma = O(t^{1-\alpha}) \rightarrow 0, \quad t^2\gamma^2 = O(t^{2-2\alpha}) \rightarrow 0, \quad t^2w^2 = O(t^{2-2\alpha}) \rightarrow 0,$$

we obtain

$$tK_{\parallel} = \frac{2d(4\alpha d^2 + o(1))}{E^{3/2}} + o(1).$$

Using $d \rightarrow d_*$ and $E \rightarrow 4d_*^2$,

$$tK_{\parallel} \rightarrow \frac{8\alpha d_*^3}{(4d_*^2)^{3/2}} = \alpha.$$

Thus both transverse and longitudinal principal curvatures have the same leading coefficient α/t . $\square$

Remark 6.14. The theorem is a leading-order statement in the fixed-displacement regime. It should not be interpreted as describing a single fixed level $\psi = c$, since the level value diverges as $t \rightarrow 0$ in this regime. The fixed-level geometry is described separately in Proposition 6.8.

7. DISCUSSION

7.1. What is generic and what is model-specific. The decomposition

$$\psi = fd^2$$

and the identities of Sections 4–5 are local statements on the smooth region $U = I^-(p) \setminus \mathcal{C}(p)$. They are therefore insensitive to the global topology of the cut locus. The generic theorem is correspondingly a theorem about regular values of $\psi|_U$. The explicit FLRW analysis adds information of a different type. The endpoint map is strictly monotone, which gives a description of the timelike region and uniqueness of the connecting maximizing geodesic. In that model the causal tilt satisfies $\Theta < 1$ everywhere in the timelike region, so every regular curvature-weighted level component is spacelike there.

7.2. Two singular asymptotic regimes. The distinction between fixed displacement and fixed level is essential. For fixed $x \neq 0$ one obtains

$$d \rightarrow d_* > 0, \quad \psi = t^{-4}d^2 \rightarrow \infty,$$

and

$$1 - \Theta^2 \sim \frac{t^{2\alpha}}{\mu_*^2}.$$

The hypersurfaces become asymptotically null in causal margin while their principal curvatures are asymptotic to the FLRW Hubble rate α/t . For fixed $c > 0$, by contrast,

$$d = \sqrt{c}t^2 \rightarrow 0, \quad \mu \rightarrow \infty,$$

and the corresponding hypersurface approaches the null boundary. The radial gap satisfies

$$H(t) - |x_c(t)| \sim \frac{ct^4}{2B_0}, \quad B_0 = \frac{T^{\alpha+1}}{\alpha+1}.$$

This is a genuinely different asymptotic geometry and demonstrates that the foliation by ψ -levels contains more information than a single scaling law.

7.3. Curvature-derived weights. For a general theorem it is preferable to assume directly that

$$f > 0, \quad f \in C^{r,\alpha},$$

and then impose a separate singularity condition. A curvature invariant may be used when it is positive and has the required regularity. Absolute values should not be inserted without checking the regularity at zeros. In the FLRW model, the Kretschmann scalar provides a positive weight for the entire range $0 < \alpha < 1$ and has the scaling t^{-4} .

7.4. Relation to gluing. A spacelike level component carries an induced Riemannian metric h and a second fundamental form K . These are the geometric data required in any subsequent matching problem. For a smooth join across a spacelike hypersurface, one must match the appropriate induced data in the Israel–Darmois sense [5]; distributional thin-shell sources appear when the extrinsic data have a jump. The present work does not construct a glued spacetime. It identifies and computes candidate spacelike hypersurfaces on which such a future matching problem could be posed.

7.5. Scope of the results. The general theorem does not assert that almost every full level avoids the cut locus. Such a statement is not true under the stated hypotheses. Instead, the paper identifies the smooth regular part and develops a complete causal and extrinsic calculus there. The FLRW model then supplies a setting in which the relevant timelike endpoint problem can be solved explicitly and the resulting level geometry can be followed into the singular regime.

Open Problem 7.1. For fixed-level hypersurfaces in the power-law FLRW family, determine the complete next-order expansion of the principal curvatures and the induced metric as $t \rightarrow 0^+$. In particular, determine whether the difference

$$K_{\parallel} - K_{\perp}$$

remains bounded, converges, or has a model-dependent divergent subleading term.

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APPENDIX A. EXPANDED DIFFERENTIAL-GEOMETRIC PROOFS

A.1. Proof of the Hessian formula. Let $u = d^2$. For any smooth scalar functions f, u ,

$$\text{Hess}(fu) = f \text{Hess } u + u \text{Hess } f + df \otimes du + du \otimes df.$$

Since

$$du = 2d \, dd,$$

we have

$$du(Y) = 2d \, g(e_0, Y).$$

Differentiating once more,

$$\text{Hess } u(Y, Z) = 2g(e_0, Y)g(e_0, Z) + 2d \, \text{Hess } d(Y, Z).$$

Also,

$$df(Y) = ag(e_0, Y) + g(X, Y).$$

Substitution gives

$$\begin{aligned} \text{Hess } \psi(Y, Z) &= 2fg(e_0, Y)g(e_0, Z) + 2fd \text{Hess } d(Y, Z) + d^2 \text{Hess } f(Y, Z) \\ &\quad + 2adg(e_0, Y)g(e_0, Z) + 2dg(X, Y)g(e_0, Z) \\ &\quad + 2adg(e_0, Y)g(e_0, Z) + 2dg(X, Z)g(e_0, Y). \end{aligned}$$

Collecting the coefficient of $g(e_0, Y)g(e_0, Z)$ gives $2(f+2ad)$, proving Lemma 5.1.

A.2. Proof of the restricted Hessian. For tangent vectors Y, Z ,

$$dAg(e_0, Y) + d^2g(X, Y) = 0,$$

and similarly for Z . Hence

$$g(e_0, Y) = -\frac{d}{A}g(X, Y), \quad g(e_0, Z) = -\frac{d}{A}g(X, Z).$$

The contribution from the e_0e_0 term is

$$2(f+2ad)\frac{d^2}{A^2}g(X, Y)g(X, Z),$$

while the mixed terms contribute

$$-\frac{4d^2}{A}g(X, Y)g(X, Z).$$

Together,

$$\frac{2d^2}{A^2}[(f+2ad) - 2A]g(X, Y)g(X, Z).$$

Since $A = ad + 2f$,

$$(f+2ad) - 2A = -3f,$$

so the coefficient is

$$-\frac{6fd^2}{A^2}.$$

This proves Theorem 5.2.

A.3. Derivation of the FLRW endpoint map. The Lagrangian for timelike geodesics is

$$L = \frac{1}{2}(-\dot{t}^2 + t^{2\alpha}\dot{x}^2).$$

The Euler–Lagrange equation in x gives

$$\frac{d}{d\lambda}(t^{2\alpha}\dot{x}) = 0,$$

so

$$t^{2\alpha}\dot{x} = \mu.$$

The proper-time normalization gives

$$-\dot{t}^2 + \mu^2 t^{-2\alpha} = -1,$$

therefore

$$\dot{t} = \sqrt{1 + \mu^2 t^{-2\alpha}}.$$

Hence

$$\frac{dx}{dt} = \frac{\mu t^{-2\alpha}}{\sqrt{1 + \mu^2 t^{-2\alpha}}} = \frac{\mu t^{-\alpha}}{\sqrt{t^{2\alpha} + \mu^2}},$$

and integrating gives (6). The proper-time integral gives (7).

A.4. Differential of the endpoint map and the longitudinal Hessian.

Differentiate

$$x = - \int_t^T \frac{\mu s^{-\alpha}}{\sqrt{s^{2\alpha} + \mu^2}} ds$$

with respect to μ at fixed t . One obtains

$$x_\mu = - \int_t^T \frac{s^\alpha}{(s^{2\alpha} + \mu^2)^{3/2}} ds = -J.$$

Since $J > 0$,

$$\mu_x = -\frac{1}{J}.$$

On the other hand, differentiation of the distance with respect to x using the envelope property of the maximizing geodesic gives

$$d_x = \mu.$$

Thus

$$d_{xx} = \mu_x = -\frac{1}{J}.$$

Using

$$\Gamma_{xx}^t = \alpha t^{2\alpha-1}, \quad \Gamma_{tx}^x = \frac{\alpha}{t},$$

one finds

$$\begin{aligned} \text{Hess } d(\partial_t, \partial_x) &= d_{tx} - \frac{\alpha}{t} d_x, \\ \text{Hess } d(\partial_x, \partial_x) &= d_{xx} + \alpha t^{2\alpha-1} d_t. \end{aligned}$$

Contracting with

$$e_1 = w\partial_t + \gamma t^{-\alpha}\partial_x$$

and simplifying yields

$$L_{\parallel} = \frac{\alpha}{t\gamma} - \frac{t^{-2\alpha}}{J\gamma^2}.$$

A.5. Derivation of the fixed-level asymptotics. On $\psi = c$,

$$d = \sqrt{c} t^2.$$

For large μ ,

$$\frac{s^\alpha}{\sqrt{s^{2\alpha} + \mu^2}} = \frac{s^\alpha}{\mu} + O\left(\frac{s^{3\alpha}}{\mu^3}\right),$$

and therefore

$$d = \frac{B(t)}{\mu} + O(\mu^{-3}), \quad B(t) = \int_t^T s^\alpha ds.$$

Thus

$$\mu = \frac{B(t)}{\sqrt{c}t^2}(1 + o(1)).$$

Similarly,

$$\frac{\mu s^{-\alpha}}{\sqrt{s^{2\alpha} + \mu^2}} = s^{-\alpha} - \frac{s^\alpha}{2\mu^2} + O(\mu^{-4}s^{3\alpha}),$$

which gives

$$H(t) - x_c(t) = \frac{B(t)}{2\mu^2}(1 + o(1)) = \frac{ct^4}{2B_0}(1 + o(1)).$$

Finally,

$$d \sim \sqrt{c}t^2, \quad \gamma \sim \mu \sim \frac{B_0}{\sqrt{c}t^2},$$

so

$$2d\gamma \rightarrow 2B_0.$$

Moreover,

$$4d^2 + 4dt\gamma + t^2 = 4B_0t + o(t),$$

and therefore

$$1 - \Theta^2 = \frac{4B_0t + o(t)}{4B_0^2 + o(1)} = \frac{t}{B_0}(1 + o(1)).$$

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